

Project Proposal & Problem Statement

Project Title

AURA-EMS: Context-Aware Campus Energy Demand Forecasting Framework using Heterogeneous Time-Series Fusion

1. The Problem Context

Modern educational institutions like Alliance University operate as complex micro-cities. Energy consumption profiles within the campus are highly dynamic and deeply fragmented. Academic blocks experience heavy daytime loads, while residential hostels experience sharp early-morning and late-night spikes.

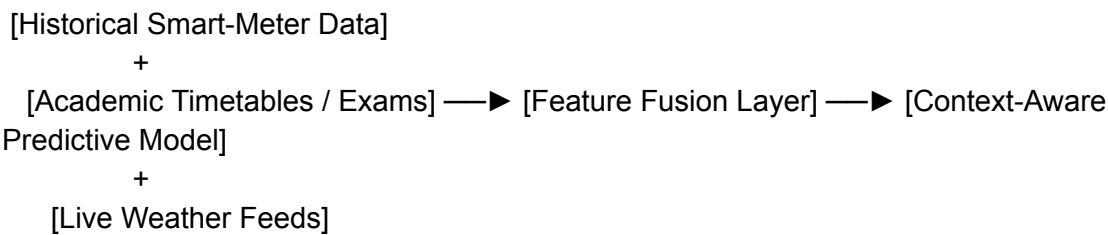
Standard energy management frameworks analyze power consumption as an isolated variable. They rely exclusively on past historical billing data. This reactive approach fails to anticipate tomorrow's power needs. It leaves the university highly vulnerable to high-tariff peak-load penalties imposed by BESCOM.

2. Comprehensive Problem Statement

"How can Alliance University transition from reactive energy tracking to proactive, context-aware demand forecasting using advanced Data Science methodologies?"

Traditional time-series models fail in a campus environment because they ignore the non-electrical, external operational catalysts that dictate actual energy usage. Power spikes and drops are heavily driven by the university timetable, exam schedules, student holidays, and sudden weather shifts.

The objective of **AURA-EMS** is to solve this by building a predictive machine learning framework. This system will fuse raw smart-meter time-series data with exogenous institutional datasets (academic calendars and local weather feeds). It will generate highly accurate, contextual, short-term energy demand forecasts, allowing the university administration to strategically mitigate peak-load costs and eliminate operational waste." [2, 3, 4]



3. Proposed Methodology & Core Modules

Our 3-member team will implement the project through three streamlined data science stages:

- **Module 1: Multi-Source Data Ingestion & Engineering:** Construct a tabular data pipeline to clean and align historical campus smart-meter logs (kW, kVA) alongside exogenous tabular features: Day-of-Week, Academic vs. Vacation status, Exam Cycles, and hourly Temperature/Humidity data.
- **Module 2: Contextual Time-Series Machine Learning:** Train and optimize supervised ML models (such as XGBoost Regressors and LightGBM) configured specifically to process exogenous variables. The models will output rolling 24-hour and 7-day predictive power consumption curves for individual campus zones.
- **Module 3: Institutional Forecast Visualizer:** Build an intuitive, interactive executive dashboard using Streamlit. This portal will display predicted vs. actual consumption, highlight impending peak-demand risks, and provide data-driven insights for campus facilities management.

4. Expected Institutional Impact (Value Proposition)

- **Peak Demand Mitigation:** Allows administrative teams to view impending load spikes 24 hours in advance, enabling manual load balancing to avoid hefty BESCOM contract penalties.
- **Resource Allocation:** Provides empirical data insights to help estate managers optimize HVAC and lighting schedules based on predicted department-level usage.
- **Academic Excellence:** Delivers a functional, production-ready data science project that demonstrates the immediate, practical application of predictive analytics on the